

Presentation vF Calculation Cheat Sheet

This sheet follows the actual presentation order from slide 13 onward. Use it as a Q&A guide: each section lists likely questions, the short answer, the formula or calculation, and the appendix slide where the full backup appears.

Slide 13: Quantitative Research Questions

Likely question: What is the quantitative component actually testing?

Short answer: It tests how LLM-mediated discovery changes marketplace outcomes across four linked dimensions: matching, monetization, paid bias, and trust.

The four quantitative questions:

Experiment 1: Does neutral LLM discovery improve consumer outcomes relative to direct platform search?
Experiment 2: How do CPC, CPA, and CPFI payment rules change welfare, revenue, and concentration?
Experiment 3: How much sponsored ranking bias can the market tolerate?
Experiment 4: Does consumer trust affect adoption, match quality, or market power?

Main metrics:

Consumers: intent fulfilment, utility, price, delivery time, conversion
Platforms: net revenue, order shares, net vs no-LLM baseline
Restaurants: profit/order, exposure concentration
LLM: revenue, click-through, shortlist relevance

Appendix reference: Slide 54 / Appx T for aggregate metric definitions; slides 55-58 / Appx U-X for experiment designs.

Slide 14: Four Actors, Two Discovery Worlds

Likely question: Where do actor attributes come from?

Short answer: The simulation draws restaurants, offers, and consumers from configured distributions, while platforms are fixed archetypes.

Platform primitives:

QuickEats: commission 20%, delivery fee \$1.50-\$4.00, logistics 10-25 min
FoodRush: commission 25%, delivery fee \$2.00-\$4.50, logistics 15-30 min
NomNom: commission 15%, delivery fee \$1.00-\$3.50, logistics 20-35 min

Restaurant attributes:

cuisine ~ Uniform(9 cuisines)
quality ~ Uniform(2.0, 5.0), rounded to nearest half-star
menu_price = quality-correlated price, clipped to \$19-\$50
prep_time ~ IntegerUniform(5, 25)
cost_ratio ~ Uniform(0.30, 0.60)
platform listings: each restaurant joins 1-3 platforms

Consumer attributes:

preferred cuisines: 2-4 cuisines
budget ~ Uniform(22, 60)
max_delivery_time ~ IntegerUniform(20, 60)
price_sensitivity = abs(Normal(1.0, 0.3))
time_sensitivity = abs(Normal(0.8, 0.3))
quality_sensitivity = abs(Normal(1.0, 0.3))
llm_trust ~ Uniform(0, 1)
platform_loyalty_i ~ Uniform(0, 1)

Appendix reference: Slide 36 / Appx B, slide 37 / Appx C, slide 39 / Appx E.

Slide 15: Simulation Pipeline

Likely question: What happens in one simulation run?

Short answer: Each run creates the market, samples consumer intents, builds each consumer's visible choice set, simulates choices, records order outcomes, and aggregates metrics.

Pipeline:

1. Generate platforms, restaurants, offers, consumers
2. For each consumer, sample one meal intent
3. Build visible choice set depending on world
4. Rank visible offers
5. Simulate consumer choice
6. Record consumer, restaurant, platform, and LLM outcomes
7. Repeat across 10 Monte Carlo runs

Meal intent:

```
intent_cuisine ~ RandomChoice(consumer.preferred_cuisines)
intent_budget = consumer.budget * Uniform(0.9, 1.0)
intent_max_time = consumer.max_delivery_time * Uniform(0.9, 1.1)
```

Appendix reference: Slide 35 / Appx A and slide 40 / Appx F.

Slide 16: Two Search Worlds

Likely question: What exactly differs between no-LLM and LLM worlds?

Short answer: The no-LLM world is fragmented platform search; the LLM world is cross-platform aggregated search.

No-LLM choice set:

```
consumer checks 2 highest-loyalty platforms
each platform applies cuisine filter
each platform returns top 5 cuisine-matching offers
max visible offers = 2 platforms * 5 offers = 10 offers
```

LLM choice set:

```
LLM sees all offers across all platforms
LLM scores every offer
LLM returns top 5 shortlist
consumer clicks or ignores, then purchases or abandons
```

Appendix reference: Slide 41 / Appx G and slide 43 / Appx I.

Slide 17: Four-Sided Outcome Scoring

Likely question: How do you score each actor's outcome?

Consumer:

```
conversion_rate = completed_orders / total_sessions
avg_intent_fulfillment = mean(intent_fulfillment | purchase)
avg_utility = mean(consumer_utility | all sessions)
avg_total_price = mean(total_price | purchase)
avg_delivery_time = mean(delivery_time | purchase)
```

Platform:

```
platform_net_revenue = sum(platform_net)
platform_order_share_i = orders_on_platform_i / total_orders
```

Restaurant:

```
avg_restaurant_profit = mean(restaurant_profit | purchase)
exposure_gini = Gini(restaurant exposure counts)
order_gini = Gini(restaurant order counts)
exposure_HHI = sum(exposure_share_i^2)
```

LLM:

```
click_through_rate = clicked_sessions / total_sessions
total_llm_revenue = sum(llm_payment)
avg_shortlist_relevance = mean(semantic match of shortlist offers)
```

Appendix reference: Slide 54 / Appx T.

Slide 18: Intent Fulfilment

Likely question: What is intent fulfilment, exactly?

Short answer: It is the objective task-success score for a completed order: did the order satisfy the stated cuisine, budget, time, and quality requirements?

Formula:

```
intent_fulfilment =
cuisine_factor * price_factor * time_factor * quality_factor
```

Factors:

```
cuisine_factor = 1 if offer_cuisine == intent_cuisine else 0

price_factor = 1 if total_price <= budget
price_factor = max(0, 1 - 0.5 * (total_price - budget) / budget) otherwise

time_factor = 1 if delivery_time <= max_time
time_factor = max(0, 1 - 0.02 * (delivery_time - max_time)) otherwise

normalized_quality = (quality - 2.0) / 3.0
quality_factor = 0.60 + 0.40 * normalized_quality
```

Appendix reference: Slide 46 / Appx L.

Slides 19-20: Concrete Walkthrough

Likely question: How does ranking influence final choice?

Short answer: Ranking changes the choice set. Consumer choice then occurs probabilistically over the visible offers.

No-LLM path:

```
Intent: Korean, budget $50.14, max time 45 min
Consumer checks FoodRush + NomNom because those are top loyalty platforms
Platforms show cuisine-filtered platform-ranked offers
Consumer chooses among visible offers + no-purchase option
```

LLM path:

```
LLM sees all platform offers
LLM returns top-5 shortlist
Stage 1: consumer clicks one recommendation or no-clicks
Stage 2: if clicked, consumer purchases or abandons
```

Choice rule:

$$P(\text{choice} = i) = \exp(U_i) / \sum_j \exp(U_j)$$

No-LLM outside option:

U_no_purchase = 2.5

LLM outside options:

U_no_click = 1.2
U_abandon_after_click = 0.6

Walkthrough result:

No-LLM: NomNom, 51 min, intent fulfilment 0.763
LLM Neutral: NomNom, 29 min, intent fulfilment 0.917
CPC: FoodRush, 42 min, intent fulfilment 0.866
CPFI: QuickEats, 20 min, intent fulfilment 0.931

Appendix reference: Slide 48 / Appx N and slide 49 / Appx O.

Slide 21: Experiment Roadmap

Likely question: Why these four experiments?

Short answer: They build a chain of logic.

Exp 1 establishes the baseline consumer-platform tension.
Exp 2 tests payment mechanisms once monetization enters.
Exp 3 tests sensitivity to paid ranking strength.
Exp 4 separates algorithmic match quality from consumer adoption/trust.

Appendix reference: Slides 55-58 / Appx U-X.

Slide 22: Experiment 1 Consumer Results

Likely question: Why does neutral LLM improve consumer matching?

Short answer: It aggregates all platforms and ranks by intent fit rather than platform-specific revenue-aware feeds.

Main results:

Intent fulfilment: 0.683 -> 0.747
Conversion: 80.4% -> 81.6%
Avg price paid: \$34.19 -> \$32.48
Delivery time: 36.4 min -> 34.7 min

Mechanism:

No-LLM: consumers see up to 10 offers from 2 loyalty-selected platforms
LLM Neutral: consumers see top 5 offers from all platforms

Appendix reference: Slide 55 / Appx U.

Slide 23: Experiment 1 Platform Results

Likely question: Why does neutral LLM hurt platforms?

Short answer: Neutral LLM optimizes consumer fit, not platform commission revenue or sponsored platform placement.

Result:

No-LLM platform net: \$69,366
LLM Neutral platform net: \$67,884
Difference: -\$1,483

Platform net formula:

```
commission = menu_price * platform_commission_rate
subsidy_spend = 0.5 * promo_discount
platform_net = commission + delivery_fee - subsidy_spend - llm_payment
```

Precise framing:

Neutral LLM is not privately attractive to platforms because it reduces aggregate platform net.
This motivates sponsored or preferential LLM integrations, not "paying for neutral access."

Appendix reference: Slide 50 / Appx P and slide 55 / Appx U.

Slide 24: Why Paid Regimes Can Restore Incentives

Likely question: If platforms pay the LLM, how can they become more profitable?

Short answer: Payment is a cost, but paid routing can redirect more orders to sponsors, increasing commission and delivery revenue.

Core logic:

```
extra platform revenue from redirected orders - LLM payment = net gain/loss
```

Concrete full-run example for CPC:

```
No-LLM platform net: $69,366
CPC gross platform net with routing bias but zero LLM payment: about $75,569
Gross surplus: about $6,203
LLM captures 50%: about $3,101
Final CPC platform net: $72,468
```

Appendix reference: Slide 53 / Appx S and slide 50 / Appx P.

Slide 25: Payment Mechanisms

Likely question: What do CPC, CPA, and CPFI mean mechanically?

Payment triggers:

```
Neutral: LLM payment = 0
CPC: pay if user clicks an LLM recommendation
CPA: pay if user completes an order
CPFI: pay if user completes an order, scaled by intent fulfilment
```

Payment formulas:

```
CPC: llm_payment = nominal_CPC_rate * payment_scale if clicked
CPA: llm_payment = nominal_CPA_rate * payment_scale if purchased
CPFI: llm_payment = nominal_CPFI_rate * intent_fulfilment * payment_scale if purchased
```

Appendix reference: Slide 52 / Appx R and slide 49 / Appx O.

Slide 26: Experiment 2 Consumer Welfare

Likely question: Why does CPFI preserve fit better than CPC?

Short answer: CPC/CPA paid boosts depend on platform sponsorship rates; CPFI multiplies the boost by predicted intent fulfilment.

LLM ranking:

```
llm_score =
0.35 * semantic_match
+ 0.20 * affordability_fit
```

```

+ 0.28 * quality / 5
- 0.12 * delivery_time / 90
- 0.05 * total_price / 40
+ sponsorship_bias

```

CPC and CPA sponsorship bias:

```

sponsorship_bias = bias_strength * platform_rate / max_platform_rate

```

CPFI sponsorship bias:

```

sponsorship_bias =
bias_strength * platform_rate / max_platform_rate * predicted_intent_fulfilment

```

Results:

```

Intent fulfilment:
Neutral 0.747
CPC      0.716
CPA      0.726
CPFI     0.747

```

Appendix reference: Slide 44 / Appx J and slide 45 / Appx K.

Slide 27: Experiment 2 Platform Revenue

Likely question: Are paid regimes actually positive relative to no-LLM?

Short answer: Yes in aggregate, after WTP payment scaling.

Platform net revenue:

```

No-LLM:      $69,366
LLM Neutral: $67,884
LLM CPC:     $72,468
LLM CPA:     $72,239
LLM CPFI:    $71,309

```

Net vs no-LLM:

```

Neutral: -$1,483
CPC:     +$3,101
CPA:     +$2,873
CPFI:    +$1,942

```

Important nuance:

```

Aggregate platform net can rise while some platforms lose traffic.
This is a competitive sponsored-discovery game, not a cooperative all-platform agreement.

```

Appendix reference: Slide 53 / Appx S and slide 60 / Appx Z.

Slide 28: Experiment 2 Market Concentration

Likely question: What does competitive externality mean?

Short answer: One platform's sponsorship redirects consumer attention away from rivals, harming them even if they did not pay the LLM.

FoodRush order share:

```

Neutral: 32.6%
CPC:     66.7%
CPA:     60.4%
CPFI:    46.5%

```

NomNom order share:

```
Neutral: 26.9%
CPC:    3.8%
CPA:    3.6%
CPFI:   12.8%
```

Competitive externality:

FoodRush sponsors LLM -> FoodRush rises in shortlist -> more FoodRush orders -> NomNom loses traffic

Appendix reference: Slide 60 / Appx Z.

Slide 29: Experiment 3 Bias Sweep

Likely question: What is bias strength, and how is it different from sponsorship bias?

Short answer: Bias strength is the global knob; sponsorship bias is the actual ranking boost for an offer.

Definitions:

```
bias_strength = global intensity parameter
platform_rate = platform-specific nominal sponsorship rate
sponsorship_bias = additive score boost applied to a specific offer
```

For CPC/CPA:

```
sponsorship_bias = bias_strength * platform_rate / max_platform_rate
```

For CPFI:

```
sponsorship_bias =
bias_strength * platform_rate / max_platform_rate * predicted_intent_fulfilment
```

Bias sweep results:

```
bias_strength 0.0: intent fulfilment 0.747
bias_strength 0.3: intent fulfilment 0.716
bias_strength 1.0: intent fulfilment 0.649
No-LLM baseline:  intent fulfilment 0.683
```

Appendix reference: Slide 57 / Appx W and slide 45 / Appx K.

Slide 30: Experiment 4 Trust Sweep

Likely question: Why does trust affect conversion but not match quality?

Short answer: Trust enters consumer utility, not the LLM ranking function. It affects whether consumers click/purchase, not which offers appear.

Utility term:

```
LLM trust bonus = 0.6 * llm_trust * cuisine_match
```

Trust sweep:

```
LLM conversion rises from 78.6% to 85.2%
Intent fulfilment stays roughly 0.747 -> 0.749
```

Interpretation:

```
trust affects action/adoption
ranking algorithm affects match quality
```

Appendix reference: Slide 47 / Appx M and slide 58 / Appx X.

Slide 31: Main Takeaways

Likely question: What is the defensible final interpretation?

Answer:

1. Neutral LLM improves consumer matching but lowers aggregate platform net.
2. Paid regimes restore sponsor-side incentives by redirecting traffic.
3. CPC over-rewards attention and creates concentration.
4. CPFI is the preferred mechanism because paid amplification is conditional on fulfilled intent.
5. Bias caps around 0.1-0.2 preserve most consumer gains while allowing monetization.

Appendix reference: Slide 60 / Appx Z for participation nuance.

Slide 32: Limitations

Likely question: What are the model's biggest limitations?

Concise answers:

Static one-shot model: no repeated adaptation.
No platform opt-out: all platform offers remain visible to the LLM.
Sponsored boost is ranking-only: no explicit restaurant-to-platform ad cash flow.
Passive restaurants: no price, promo, or platform-listing response.
Hand-coded cuisine similarity: no embedding-based semantic model.
Single LLM intermediary: no LLM competition.
No courier capacity constraints.
Fixed trust: no consumer learning over time.

Appendix reference: Slide 59 / Appx Y.

Slide 33: Final Conclusion

Likely question: What should someone remember?

One-sentence answer:

LLM intermediation is not automatically good or bad; welfare depends on what the LLM is paid to optimize.

Recommended design:

Cost Per Fulfilled Intent + explicit cap on paid ranking bias

Why:

CPFI ties LLM monetization to consumer task success.
Bias caps prevent paid placement from overwhelming relevance.

Appendix reference: Slide 60 / Appx Z.

Appendix Slide Map

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